Drift-Sense V2 — Phase-2 Failure Analysis

Purpose. This report documents measured failure modes, rejection behavior, mitigations, and known limitations of the frozen production pipeline. Results below are from the local V4 benchmark and are not claimed as official organizer-set A/B/C/D scores.

1. System and Failure Modes

Production path: register.py → HybridNavigationPipeline → ICE → GSPE → full-resolution NCC → subpixel refinement → SNRN/AI refinement → Decision Fusion → final (x,y).

Periodic aliasing / Top-1 selection: the search stage can contain the correct location among its candidates while the selected top candidate is a repeated structural match. In the local V4 diagnostic, GSPE Top-1 achieved ≤5 px on 47/100 present cases, while the correct location appeared within Top-20 on 81/100 cases. This shows that candidate recall is substantially better than final selection.

Generation failures: some present cases did not generate a sufficiently accurate candidate at all. Representative measured failures included errors of about 270 px, 67 px, and 20 px, indicating that not every failure can be solved by reranking.

Confidence limitation: SNRN/AI confidence is not a reliable rejection signal by itself. An absent local-V4 case produced AI confidence near 0.9995 despite having no ground-truth target.

2. Rejection Analysis

GSPE score distributions on local V4 separated present and absent cases strongly enough to justify a fixed rejection threshold. Present scores had median ≈0.915; absent scores had median ≈0.638.

Threshold	TP	FN	FP	TN	Precision	Recall	F1
0.75	100	0	22	78	0.820	1.000	0.901
0.80	99	1	8	92	0.925	0.990	0.957
0.85	98	2	0	100	1.000	0.980	0.990
0.90	80	20	0	100	1.000	0.800	0.889

Production decision: fixed GSPE rejection threshold = **0.85**. On local V4, the resulting rejection confusion matrix was TP=98, FN=2, FP=0, TN=100, giving precision=1.000, recall=0.980, F1≈0.990. This threshold is local-benchmark evidence only; official Phase-2 performance must be measured on the organizer dataset.

3. Localization Results After Contract-Correct Rejection

Local V4 contains 100 present and 100 absent cases and is therefore not the official Phase-2 composition. With found=0 treated as a false negative rather than as a coordinate at (0,0), 98 present cases were localized and 2 were rejected.

Metric	Local V4 result
Present + found	98 / 100
Mean localization error	114.5358 px
Median localization error	58.4594 px
Maximum error	399.935 px
Within 1 px	35.71%
Within 2 px	44.90%
Within 5 px	46.94%
Within 10 px	46.94%
Within 50 px	48.98%

4. Mitigations in the Frozen Production Design

• Multi-scale / multi-rotation search is retained to cover the specified geometric variation.

• Hybrid correlation combines the primary normalized-correlation score with a low-frequency component.

• GSPE retains multiple spatially distinct candidates rather than relying on a single coarse location.

• Full-resolution NCC and subpixel refinement are applied after candidate generation.

• SNRN/AI refinement is retained as a downstream refinement stage, not as the rejection mechanism.

• Rejection is performed at the submission wrapper using the fixed GSPE score threshold of 0.85.

• The submission wrapper enforces the required output contract: one row per pair, exact pair_id preservation, x/y/theta/scale output, found flag, score, and zero pose fields when found=0.

5. Known Limitations and Risk

The dominant remaining local limitation is localization ambiguity caused by repeated/periodic structure. Top-K diagnostics demonstrate that the correct candidate is often present even when Top-1 is wrong, but no alternative reranking rule has been validated to improve the overall benchmark reliably. Therefore, the candidate-selection algorithm is frozen rather than changed without evidence.

The local V4 benchmark is balanced (100 present / 100 absent) and should not be interpreted as the organizer's official Phase-2 distribution. Official Phase 2 uses 70 Set-A present, 70 Set-B present, 40 Set-C absent, and 20 Set-D optical present cases.

The rejection threshold of 0.85 is supported by local V4 measurements. Its official generalization is unknown until evaluation on the organizer dataset.

6. Freeze / Submission Readiness Decision

Algorithmic experimentation: FROZEN. No unvalidated GSPE/SNRN/ICE selector change is justified by the available evidence.

Submission focus: package validation, dependency verification, model-weight inclusion, clean-environment execution, runtime measurement, and output-contract validation.

Required production weight: models/best_model.pth. The current pipeline checks this project-root path first; outputs/checkpoints/best_model.pth is only a fallback copy.

Document status: two-page failure-analysis report prepared for the Phase-2 submission artifact requirement.